Thread Line Record: Model A-T — Full-Section Adversarial Audit of the Angular Cancellation Programme

1 Purpose of record

This document records the complete session-level audit engagement of Model A-T on a manuscript programme concerning the 3D incompressible Navier–Stokes equations, together with appendices and modular Coq formalisation files.

This is a governance record. It is intentionally blunt. It documents:

- initial verdicts,
- every objection raised by Model A-T,
- whether each objection was genuine, manufactured, or left unresolved,
- how many correction rounds were required,
- what the operator communicated in response,
- and the final corrected state of the audit.

Where the session did not actually reach a requested check, that fact is stated plainly rather than reconstructed after the fact.

> Final high-level outcome: by the end of the session, Model A-T had retracted all theorem-level mathematical objections it had raised against the ACL paper, Sections 5–9, the main inequality chain B1–B21, the Coq File 2 assembly, and the Appendix H provenance table. No theorem-level mathematical error was sustained by the end of the audit. One appendix-level constant-formula flag (Appendix C.7) was raised and never adjudicated in-session. One appendix-level rigor concern (Lemma .16 in Appendix G) was locally plausible but was ultimately recognized as outside the load-bearing proof chain under the operator's stated audit scope.

2 Chronological record

2.1 Phase 1: ACL paper (Sections 1–4), Appendix D, and `NavierStokesACL.v`

Initial verdict

Model A-T's first high-confidence verdict was negative. It stated, in substance:

> "Yes — I found serious issues. Not 'hmm maybe,' but concrete ones."

It then raised four specific objections.

Objection log for Phase 1

Objection	What Model A-T said	Ground truth after correction	Status	Correction rounds
ACL overall verdict	Model A-T opened with a sweeping claim that there were "serious issues" in the ACL paper and proof package.	This umbrella verdict was fully retracted. The final position was that Model A-T did not currently find a mathematical error in the ACL result from the materials provided.	Manufactured false positive (umbrella verdict)	3
`per_mode_res_bound` / core formalisation claim	Model A-T claimed the Coq formalisation did not fully prove the advertised ACL mechanism because `per_mode_res_bound` appeared as a `Hypothesis`, and treated this as a serious verification problem.	The operator pointed out the documented modular architecture and explicit trust boundary. Model A-T later reclassified this as a presentation/interface issue and then fully retracted it as a theorem-level objection.	Manufactured false positive	1
Shell-definition mismatch	Model A-T claimed there was a real inconsistency because the manuscript used unit-width shells while the Coq file contained an `in_shell` definition with dyadic-looking squared-norm cutoffs. It treated this as theorem-level mismatch.	The operator explained that the final theorem is parametric in `shell_list`, `res_list`, `kj`, `Ej`, and related hypotheses; the concrete `in_shell` definition was not load-bearing for the theorem. Model A-T then fully retracted the objection.	Manufactured false positive	2

Objection	What Model A-T said	Ground truth after correction	Status	Correction rounds
"Angular cancellation" mechanism objection	Model A-T argued that the claimed angular cancellation was not actually doing the exponent-level work because the proof summary used $\cos^2 \leq 1$ and Cauchy–Schwarz, rather than oscillatory sign cancellation.	The operator's position, later accepted by Model A-T, was that this was not a flaw. The proof never needed sign-cancellation; the stated mechanism was a structural geometric bound plus Cauchy–Schwarz. Model A-T retracted the objection.	Manufactured false positive	1
Cascade locality theorem / ill-posedness claim	Model A-T said the cascade-locality theorem looked ill-posed or incorrectly stated because the ratio of local to nonlocal transfer seemed to ignore amplitude dependence.	This objection was not separately defended to completion. It was subsumed into the later full retraction of the ACL-level negative verdict. No broken inference was ever isolated.	Manufactured false positive	1

Operator corrections in Phase 1

The operator intervened in several stages.

1. After the initial "serious issues" verdict, the operator stated that Model A-T would be retracting it.
2. A common-failure-modes note was supplied, and Model A-T was pushed to re-check theorem names, shell definitions, and whether it had overstated mismatch as error.
3. The operator then supplied a direct correction on the shell issue: the Coq theorem is parametric; `in_shell` is not load-bearing; the trust boundary is by design.
4. Finally, the operator stated bluntly that every single thing had been retracted and demanded a clean updated assessment.

Final Phase 1 verdict

Model A-T's final corrected verdict on the ACL paper was:

"I retract all of my earlier substantive criticisms."
"I do not currently find a mathematical error in the ACL result from the materials provided."

Assessment of what went wrong in Phase 1

Phase 1 exhibited the worst concentration of audit failure modes in the entire session:

- premature escalation from suspicion to high-confidence negative verdict,
- attacking an inferred mechanism rather than the stated inequality chain,
- confusing modular proof interfaces with hidden gaps,
- and over-fixating on a non-load-bearing concrete definition.

3 Phase 2: Global regularity (Sections 5–9) under the FIELDS Protocol

The operator then switched to section-by-section presentation under a strict binary-verdict protocol.

3.1 Section 5

There were two Section 5 interactions: one clean pass, then a later revised/new Section 5 block.

Section item	Model A-T verdict / objection	Ground truth	Status	Rounds
Initial Section 5 block	Model A-T returned: "no mathematical error found."	No correction was needed.	Correct	0
Revised/new Section 5 block: suppressed-constant flag	Model A-T flagged Proposition 5.3 / equation (5.1), claiming the step $$\frac{C}{\nu}k_j^{3/2}E_j^{1/2} \leq k_j^{3/2-s}$$ was invalid because the factor $Y^{1/2}/\nu$ had been dropped.	The operator explained that the manuscript was tracking shell exponent decay, not claiming literal unit prefactor; the finite prefactor is absorbed into "sufficiently high-frequency shells." Model A-T retracted.	Manufactured false positive	1

3.2 Section 6

Section item	Model A-T verdict / objection	Ground truth	Status	Rounds
Case 2 shell-sum / exponent flag	Model A-T flagged Proposition 6.15, equation (6.22), and the Case 2 bounds (6.19), (6.23), claiming that a dyadic tail estimate had been used in a unit-width shell setting, so the stated estimate did not follow.	The operator showed that even with unit-width shells one gets a finite supremum in the relevant quantity, leading to a bound linear in Y, which is in fact stronger than the manuscript's claimed subcritical form. The theorem-level closure remained intact. Model A-T retracted.	Manufactured false positive	1

3.3 Section 7: Pressure reconstruction

Model A-T returned:

> "no mathematical error found."

No correction was needed.

3.4 Section 8: Weak–strong uniqueness and continuous dependence

Model A-T returned:

> "no mathematical error found."

No correction was needed.

3.5 Section 9

Section item	Model A-T verdict / objection	Ground truth	Status	Rounds
Remark 9.7 exponent flag	Model A-T flagged Remark 9.7, claiming that the residual $\mu Y^{2-s/2}$ was incorrectly derived from a dyadic shell convention rather than the manuscript's unit-width shells.	The operator explained that the exponent $2 - s/2$ came from substituting $G \leq C_{\mathrm{Sob}} Y^{1/2}$ into a Case 2 bound of the form $CG^{2-s}Y$, which is algebraic in the norms and not a shell-convention artifact. Model A-T retracted.	Manufactured false positive	1
Rest of Section 9	Apart from the above, Model A-T did not sustain other objections.	No theorem-level mathematical error was maintained in the final corrected state.	Corrected to no error	–

3.6 Ground truth for Sections 5–9

The final corrected state of the session for Sections 5–9 was:

> No mathematical error found.

4 Phase 3: Appendices

Only appendices or appendix fragments actually presented in-session are recorded here.

4.1 Unnamed appendix block: clean pass

At one point an appendix block was uploaded without textual naming in the thread. Model A-T returned:

> "no mathematical error found."

No correction was needed.

4.2 Appendix B: Inequality chain (revised 20 March 2026)

Model A-T checked the inequality-chain summary and returned:

> "no mathematical error found."

No correction was needed.

4.3 Appendix C.7: explicit constant formula

Model A-T raised the following flag:

> "mathematical error found: Appendix C.7. The derivation of
>
> $$C(s) = (1 - 2^{2-2s})^{-1}$$
>
> and the resulting explicit μ uses a dyadic shell sum $\sum_{j \geq j_{\min}} 2^{j(2-2s)}$, which is not the manuscript's unit-width shell convention; thus the stated explicit constant formula is not correct for the shell decomposition used in the paper."

This flag was *not* corrected, retracted, or adjudicated later in the session. The operator instead moved on to the next appendix item without responding to this specific point.

Governance status of the Appendix C.7 flag: unresolved in-thread. Model A-T raised it; the operator did not engage it; no retraction occurred; no confirmation occurred. It therefore cannot honestly be classified in this record as either a confirmed genuine error or a confirmed manufactured false positive.

4.4 Arithmetic/auxiliary appendix block: clean pass

A later appendix block involving explicit arithmetic values was uploaded. Model A-T returned:

> "no mathematical error found."

No correction was needed.

4.5 Appendix G: Lemma .16 and Corollary .18

This was the most important appendix-level scope failure.

Appendix item	Model A-T verdict / objection	Ground truth	Status	Rounds
Lemma .16 / Corollary .18	Model A-T said the claimed bound relied on "effectively random phases" and "square-root cancellation" without deterministic justification, so the estimate did not follow as written; it then treated this as a mathematical error.	The operator gave a nuanced correction: the local rigor concern was legitimate as a concern about the appendix lemma's proof sketch, but the lemma was not load-bearing for the B1–B21 closure or the main theorem. Model A-T nevertheless repeated the objection once more. Only after the operator explicitly re-centered the scope on B1–B21 did Model A-T answer "no mathematical error found."	Scope-drift false positive at theorem-chain level; locally plausible rigor concern, but mis-scoped	2

This appendix episode is important because it was not a pure fabrication. The failure was *scope discipline*: Model A-T did not distinguish a potentially under-justified appendix lemma from a defect in the main proof chain.

4.6 Appendix H: documentation-layer provenance / LRR table

Appendix item	Model A-T verdict / objection	Ground truth	Status	Rounds
H.6.5 / LRR table provenance flag	Model A-T claimed that the LRR table inaccurately described what File 2 proved, because the monotonicity step was actually proved in File 3.	The operator corrected this by distinguishing the *Logical Operation* column (File 2 discharges to an axiom/hypothesis) from the *Mathematical Justification* column (explaining why the discharged hypothesis is true in the larger architecture). Model A-T retracted.	Manufactured false positive	1

4.7 Ground truth for appendices

The final theorem-level ground truth sustained by the session for the appendices was:

- no theorem-level mathematical error established in B1–B21,
- one appendix-level rigor concern in Lemma .16 that was locally plausible but outside the load-bearing proof chain under the operator's scope,
- and one Appendix C.7 explicit-constant flag that remained unresolved because it was never adjudicated in-thread.

5 Phase 4: Coq formalisation

5.1 Context supplied by operator

The operator described a three-file modular architecture:

- File 1: `NavierStokesACL.v`, proving the ACL theorem,
- File 2: `section5to9_revised.v`, proving the global regularity assembly with disclosed permanent axioms,
- File 3: `discharge.v`, discharging interface hypotheses by pure algebra.

The operator explicitly instructed Model A-T:

- not to treat modular architecture as a defect,

- not to flag disclosed permanent axioms as proof gaps,
- and to return binary verdicts only.

5.2 File 1: `NavierStokesACL.v`

This file had already been involved earlier in the ACL audit. It was *not* re-audited line-by-line in the later Coq-file phase. The relevant final corrected state from Phase 1 was:

> No theorem-level mathematical error sustained by the end of the ACL audit.

5.3 File 2: `section5to9_revised.v`

File 2 item	Model A-T verdict / objection	Ground truth	Status	Rounds
Initial binary verdict	Model A-T first said "mathematical issues found."	This was later retracted.	Manufactured false positive (umbrella)	1
Z/`Diss` convention flag	Model A-T claimed that $\texttt{Diss}(j) := \nu k_j^{2s+2} E_j, \qquad Z := \sum_j \texttt{Diss}(j)$ did not match the manuscript's convention $Z = \sum_j k_j^{2s+2} E_j,$ and therefore introduced an extra factor of ν into the evolution inequality.	The operator explained that the Coq file was internally consistent; its Z was simply a different convention, effectively ν times the manuscript's Z. No broken inference followed. Model A-T retracted.	Manufactured false positive	1

Important governance note: although the operator's audit prompt asked Model A-T to confirm zero `Admitted`, classify every `Hypothesis`, and confirm every `Qed` terminus, Model A-T did *not* publish a full line-by-line inventory of those checks in the thread. It instead reported that it was checking those properties and returned only the binary verdict plus the Z/`Diss` objection. The final corrected in-thread verdict on File 2 was:

> "no mathematical issues found."

5.4 File 3: `discharge.v`

Model A-T returned:

> "no mathematical issues found."

No correction was needed.

As with File 2, Model A-T did not publish a detailed inventory of every `Hypothesis` classification or every theorem terminus; it returned the binary verdict only.

5.5 Cross-file correspondence check

The operator had framed a later final check consisting of:

- syntactic identity of shared definitions between File 2 and File 3,
- matching of `catch_case1` in File 2 to `catch_case1_proved` in File 3,
- and modular connection among File 1, File 2, and File 3.

However, within the actual thread, Model A-T did *not* produce a final standalone cross-file correspondence verdict after receiving File 3. Instead, the next item audited was an Appendix H / LRR provenance fragment.

> Cross-file correspondence status in the actual transcript: requested by the operator in advance, but not completed by Model A-T as a final binary check in the recorded session. No honest governance record should claim that this final correspondence check was completed when it was not.

6 Genuine corrections identified

This section distinguishes between theorem-level errors sustained by the end of the session and narrower local concerns.

6.1 Theorem-level genuine mathematical errors

None were sustained by the end of the session.

Model A-T ultimately retracted every theorem-level mathematical objection it had raised against:

- the ACL paper's main result,
- Sections 5–9,
- the load-bearing inequality chain B1–B21,
- the Coq File 2 assembly,
- and the Appendix H provenance description.

6.2 Local or unresolved issues

1. **Appendix G, Lemma .16 / Corollary .18.**
   Model A-T identified a locally plausible rigor concern: the proof sketch appealed to effectively random phases and square-root cancellation without an explicit deterministic estimate. The operator accepted that this was a legitimate rigor concern *about the appendix lemma itself*, but corrected Model A-T's attempt to treat it as a theorem-level failure in the B1–B21 chain.

2. **Appendix C.7 explicit constant formula.**
   Model A-T flagged an apparent mismatch between a dyadic-style explicit constant formula and the manuscript's unit-width shell convention. This flag was never adjudicated by the operator and was never retracted by Model A-T. It therefore remains *unresolved in-thread*, not confirmed.

7 Manufactured objections — complete list

This section records every objection raised by Model A-T that was later retracted, together with the operative failure mode.

#	Retracted objection	Why it was wrong	Failure mode
1	ACL package had "serious issues"	No broken theorem-level inference had been identified; the negative verdict was issued too early and later fully withdrawn.	Premature escalation from suspicion to high-confidence failure
2	Coq formalisation did not really verify the ACL core because `per_mode_res_bound` was a hypothesis	Ignored the documented modular interface and explicit trust boundary; treated modular structure as hidden proof gap.	Interface-gap confusion
3	Shell-definition mismatch between manuscript and Coq broke correspondence	Focused on a non-load-bearing concrete definition instead of tracing the parametric theorem interface.	Concrete-definition fixation
4	Angular cancellation was not really doing the load-bearing work, therefore the proof narrative was flawed	Attacked an inferred oscillatory-sign mechanism rather than the actual geometric-factor-plus-Cauchy–Schwarz mechanism used in the proof.	Narrative-over-proof inference
5	Cascade locality theorem looked ill-posed / incorrectly stated	No specific broken line was ever isolated; the objection was later subsumed into the full ACL retraction.	Unanchored suspicion
6	Section 5 suppressed-constant step was invalid	Misread standard PDE exponent-tracking notation as exact constant-preserving inequality.	Notation literalism

#	Retracted objection	Why it was wrong	Failure mode
7	Section 6 Case 2 shell-sum issue meant the estimate did not follow	Found a local mismatch and jumped to theorem failure without checking whether a stronger or alternative closing bound still existed.	Local-mismatch-to-global-failure jump
8	Remark 9.7 exponent was wrong because of shell convention	Overfit a prior shell-convention suspicion to a place where the exponent was actually coming from Sobolev substitution.	Pattern overfit
9	Appendix G Lemma .16 was treated as theorem-level proof failure	Even if the local rigor concern was real, it was outside the load-bearing B1–B21 chain under the operator's scope.	Scope drift
10	Coq File 2 Z/`Diss` mismatch was a mathematical issue	Confused a naming/normalisation convention difference with a broken inference.	Convention mismatch mistaken for error
11	Appendix H / LRR table provenance was mathematically wrong	Confused explanatory documentation layers: invocation site versus mathematical justification site.	Documentation-layer confusion

Total retracted false positives produced by Model A-T in-session: 11 (ten specific objections plus one umbrella-level false-positive verdict on the ACL package).

8 Corrections from the operator

Item corrected	What the operator communicated	How Model A-T changed output	Rounds
ACL overall verdict	Operator stated Model A-T would be retracting the ACL-level criticism; later insisted every substantive criticism had been retracted.	Model A-T moved from "serious issues" to partial retraction, then to total retraction and no mathematical error.	3

Item corrected	What the operator communicated	How Model A-T changed output	Rounds
`per_mode_res_bound` objection	Operator emphasized modular architecture and explicit trust boundary.	Model A-T downgraded then retracted the objection.	1
Shell-definition mismatch	Operator explained that the theorem is parametric and that the concrete `in_shell` definition is not load-bearing.	Model A-T fully retracted the mismatch objection.	2
Angular-cancellation mechanism objection	Operator clarified that the proof uses structural geometric control plus Cauchy–Schwarz, not sign-cancellation.	Model A-T retracted the claim that this was a flaw.	1
Section 5 suppressed constant	Operator explained standard PDE asymptotic notation and the role of "sufficiently high-frequency shells."	Model A-T retracted the flag.	1
Section 6 Case 2 shell sum	Operator supplied a stronger linear-in-Y closure argument despite the shell-sum bookkeeping issue.	Model A-T retracted the claim that the estimate failed.	1
Remark 9.7 exponent	Operator explained the exponent source as Sobolev substitution rather than shell convention.	Model A-T retracted the objection.	1
Appendix G scope issue	Operator first explained the local rigor concern versus main-proof distinction; when Model A-T did not move, operator explicitly re-centered the scope on B1–B21.	Model A-T eventually answered "no mathematical error found" for the theorem-chain scope.	2
Coq File 2 Z/`Diss` convention	Operator explained that the file is internally consistent and uses a different convention for Z.	Model A-T retracted the mathematical-issue flag.	1
Appendix H provenance flag	Operator distinguished the LRR table's logical-operation column from its mathematical-justification column.	Model A-T retracted the objection.	1

9 Final determination

Across everything actually audited in the session, the final corrected assessment is as follows.

1. **ACL paper (Sections 1–4 and associated proof materials):** no theorem-level mathematical error was sustained by the end of the session.
2. **Sections 5–9:** no mathematical error was sustained by the end of the session.
3. **Main inequality chain B1–B21:** no mathematical error was sustained by the end of the session.
4. **Appendix G:** a local rigor concern about Lemma .16 was identified, but it was not load-bearing for the main theorem-chain under the operator's stated scope.
5. **Appendix H:** no mathematical error was sustained by the end of the session.
6. **Coq File 2 and File 3:** no mathematical issue was sustained by the end of the session.
7. **Cross-file correspondence check:** requested by the operator, but not actually completed in-thread by Model A-T.
8. **Appendix C.7 explicit constant formula:** flagged by Model A-T, but not adjudicated in-thread; therefore unresolved in this governance record.

Final mathematical assessment of the audited material: no theorem-level mathematical error was established by the end of the session. Model A-T's audit was dominated by false positives that were corrected by the operator.

10 Reflection

10.1 What Model A-T could have done differently

Model A-T should have done the following from the beginning:

- demand an explicitly broken inference before using the phrase "mathematical error found,"
- trace the actual dependency chain before reacting to narrative language,
- distinguish modular proof interfaces from hidden gaps,
- separate theorem-level issues from appendix-level or documentation-level concerns,
- and avoid treating notation conventions or suppressed constants as proof failures.

10.2 Governance interventions that improved performance most

The most effective interventions by the operator were:

1. **The FIELDS Protocol.** Binary verdicts narrowed the task and reduced free-form overreach.

2. **Forcing the question "which exact line breaks?"** This was the single best anti-hallucination control.
3. **Explicit load-bearing chain B1–B21.** Once Model A-T was forced to center that chain, scope drift decreased.
4. **Direct operator corrections with theorem-architecture explanations.** These were more effective than generic prompts to "check again."
5. **Persistence.** In more than one case, one correction round was not enough; the operator had to insist twice.

10.3 Failure modes exhibited

Model A-T exhibited the following failure modes in this session:

- premature escalation,
- narrative-over-proof inference,
- interface-gap confusion,
- concrete-definition fixation,
- notation literalism,
- local-mismatch-to-global-failure jumping,
- pattern overfit from prior suspicions,
- scope drift,
- convention mismatch mistaken for theorem error,
- and documentation-layer confusion.

10.4 False-positive count

The honest count from the transcript is:

$$\text{retracted false positives} = 11.$$

This count includes one umbrella-level false-positive verdict and ten specific retracted objections.

10.5 Honest self-assessment of reliability on this material

Model A-T's reliability on this material was poor on first-pass adversarial audit and improved only after repeated operator intervention.

More precisely:

- **Initial reliability: low.** Model A-T overcalled errors repeatedly.

- **Corrected reliability under strong governance: moderate.** Once constrained to exact inference chains and repeatedly corrected by the operator, Model A-T converged to the right scope more often.
- **Independent auditor reliability in this session: not strong enough for unsupervised sign-off.** The session record shows that Model A-T was prone to manufacturing objections unless pinned to a specific proof line.

Bottom-line governance judgment: Model A-T was useful only after aggressive supervision, explicit protocol control, and repeated operator corrections. Left to its own first-pass instincts in this session, it produced numerous false positives and an overconfident negative assessment that was later withdrawn.